homework2LLM_Luke_O'Neill

February 21, 2026

1 HomeWork 2

1.1 Luke O'Neill

2 1: Decoder-based Transformer

[1]:
```
%pip install torch torchvision torchaudio --index-url https://download.pytorch.
↪org/whl/cpu
```

```
Looking in indexes: https://download.pytorch.org/whl/cpu
Requirement already satisfied: torch in /opt/conda/lib/python3.11/site-packages
(2.10.0+cpu)
Requirement already satisfied: torchvision in /opt/conda/lib/python3.11/site-
packages (0.25.0+cpu)
Requirement already satisfied: torchaudio in /opt/conda/lib/python3.11/site-
packages (2.10.0+cpu)
Requirement already satisfied: filelock in /opt/conda/lib/python3.11/site-
packages (from torch) (3.20.0)
Requirement already satisfied: typing-extensions>=4.10.0 in
/opt/conda/lib/python3.11/site-packages (from torch) (4.12.2)
Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.11/site-
packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/opt/conda/lib/python3.11/site-packages (from torch) (3.3)
Requirement already satisfied: jinja2 in /opt/conda/lib/python3.11/site-packages
(from torch) (3.1.4)
Requirement already satisfied: fsspec>=0.8.5 in /opt/conda/lib/python3.11/site-
packages (from torch) (2023.6.0)
Requirement already satisfied: numpy in /opt/conda/lib/python3.11/site-packages
(from torchvision) (2.0.1)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in
/opt/conda/lib/python3.11/site-packages (from torchvision) (10.4.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/opt/conda/lib/python3.11/site-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in
/opt/conda/lib/python3.11/site-packages (from jinja2->torch) (2.1.5)
Note: you may need to restart the kernel to use updated packages.
```

[2]:
```
import torch
import torch.nn as nn
import torch.nn.functional as F
import math
import time
import numpy as np
import matplotlib.pyplot as plt
import seaborn
```

[3]:
```
class DecoderBlock(nn.Module):
    def __init__(self, d_model, num_heads, ff_hidden_dim, dropout):
        super(DecoderBlock, self).__init__()

        self.attention = nn.MultiheadAttention(d_model, num_heads , dropout =␣
 ↪dropout)
        self.norm1 = nn.LayerNorm(d_model)
        self.dropout1= nn.Dropout(dropout)
        self.linear1 = nn.Linear(d_model, ff_hidden_dim)
        self.linear2 = nn.Linear(ff_hidden_dim, d_model)
        self.norm2 = nn.LayerNorm(d_model)
        self.dropout2 = nn.Dropout(dropout)


    def forward(self, x, tgt_mask):
        attn_output, _ = self.attention(x,x,x,attn_mask = tgt_mask)
        x = x + self.dropout1(attn_output)
        x = self.norm1(x)
        ff_output = self.linear2(F.relu(self.linear1(x)))
        x = x + self.dropout2(ff_output)
        x = self.norm2(x)
        return x

class PositionalEncoding(nn.Module):
    def __init__(self, d_model, dropout = .1, max_len = 5000):
        super(PositionalEncoding, self).__init__()
        self.dropout = nn.Dropout(p=dropout)
        pe = torch.zeros(max_len, d_model)
        position = torch.arange(0, max_len, dtype = torch.float).unsqueeze(1)
        div_term = torch.exp(torch.arange(0, d_model, 2).float() * (-math.
 ↪log(10000.0) / d_model))
        pe[:, 0::2] = torch.sin(position * div_term)
        pe[:, 1::2] = torch.cos(position * div_term)
        pe = pe.unsqueeze(0).transpose(0,1)
        self.register_buffer("pe", pe)


    def forward(self, x):
```

```
        x= x + self.pe[:x.size(0),:]
        return self.dropout(x)
```

[4]:
```
def generate_square_subsequent_mask(size):
    mask = (torch.triu(torch.ones(size, size)) == 1).transpose(0,1)
    mask = mask.float().masked_fill(mask == 0, float('-inf')).masked_fill(mask␣
 ↪== 1, float(0.0))
    return mask
```

[5]:
```
class TransformerDecoder(nn.Module):
    def __init__(self, vocab_size, d_model, num_heads, ff_hidden_dim, dropout):
        super(TransformerDecoder, self).__init__()

        self.embedding = nn.Embedding(vocab_size, d_model)
        self.pos_encoder = PositionalEncoding(d_model, dropout)
        self.transformer_block = DecoderBlock(d_model, num_heads,␣
 ↪ff_hidden_dim, dropout)
        self.linear = nn.Linear(d_model, vocab_size)
        self.softmax = nn.LogSoftmax(dim = -1)

    def forward(self, x) :
        x= self.embedding(x)
        x = self.pos_encoder(x)
        tgt_mask = generate_square_subsequent_mask(x.size(0))
        x= self.transformer_block(x, tgt_mask)
        output = self.linear(x)
        output = self.softmax(output)
        return output
```

[ ]:
```
```

[6]:
```
# hyper parameters

vocab_size = 1000
d_model = 512
num_heads = 1
ff_hidden_dim = 2* d_model
dropout = .01
num_layers = 10
context_length = 50
batch_size = 1


model = TransformerDecoder(vocab_size, d_model, num_heads, ff_hidden_dim,␣
 ↪dropout)

input_tensor = torch.randint(0, vocab_size,(context_length, batch_size))
```

```
output = model(input_tensor)

predicted_indicies = output.argmax(dim=-1)
```

3 2: Total Number of HyperParameters

[7]:
```
def count_parameters(model):
    return sum(p.numel() for p in model.parameters() if p.requires_grad)

print (count_parameters(model))
```

```
3127784
```

4 3: Multiple Decoder based Transformer

[8]:
```
class MultiLayerTransformerDecoder(nn.Module):
    def __init__(self, vocab_size, d_model, num_heads, ff_hidden_dim, dropout,␣
 ↪num_layers):
        super(MultiLayerTransformerDecoder, self).__init__()

        self.embedding = nn.Embedding(vocab_size, d_model)
        self.pos_encoder = PositionalEncoding(d_model, dropout)
        self.transformer_blocks = nn.ModuleList([
            DecoderBlock(d_model, num_heads, ff_hidden_dim, dropout)
            for _ in range(num_layers)
        ])
        self.linear = nn.Linear(d_model, vocab_size)
        self.softmax = nn.LogSoftmax(dim=-1)


    def forward(self, x):
        x = self.embedding(x)
        x = self.pos_encoder(x)
        for transformer_block in self.transformer_blocks:
            tgt_mask = generate_square_subsequent_mask(x.size(0))
            x = transformer_block(x, tgt_mask)
        output = self.linear(x)
        output = self.softmax(output)
        return output
```

[9]:
```
import os
os.environ["WANDB_DISABLED"] = "true"

!pip install fsspec==2023.6.0
```

```
Requirement already satisfied: fsspec==2023.6.0 in
/opt/conda/lib/python3.11/site-packages (2023.6.0)
```

[10]:
```
# hyperParameters
vocab_size = 10000
# reducing this to keep my kernal from crashing
d_model = 512
num_heads = 4
ff_hidden_dim = 4*d_model
dropout = .1
num_layers = 10
context_size = 100
batch_size = 10

input_tensor = torch.randint(0, vocab_size, (context_length, batch_size))
model = MultiLayerTransformerDecoder(vocab_size, d_model, num_heads,␣
 ↪ff_hidden_dim, dropout, num_layers)
```

5 4: Number of HyperParameters in Multi Layer Transformer Decoder Model

[11]:
```
print(count_parameters(model))
```

```
41773840
```

6 5: Fool Me Once Paragraph

[12]:
```
input_sentence = "fool me once shame on shame on you cannot fool me again"


vocab = ['there', 'is', 'an', 'old', 'saying', 'in', 'tennessee', 'I', 'know',␣
 ↪'it', 'is', 'in', 'texas', 'probably', 'in', 'tennessee', 'that', 'says',␣
 ↪'fool', 'me', 'once', 'shame', 'on', 'shame', 'on', 'you', 'fool', 'me',␣
 ↪'you', "can't", 'get', 'fooled', 'again']
vocab = vocab[-20:]


# hyperparameters
d_model = 100
num_heads = 1
ff_hidden_dim = 4 * d_model
dropout = .1
num_layers = 4
context_length = 5
batch_size = 1
vocab_size = len(vocab)
```

```
word2id = {word: id for id, word in enumerate(vocab)}

id2word = {id:word for id, word in enumerate(vocab)}

model = MultiLayerTransformerDecoder(vocab_size, d_model, num_heads,␣
 ↪ff_hidden_dim, dropout, num_layers)


sequence = input_sentence.split()[:context_length]

input_tensor = torch.tensor([[word2id[word] for word in sequence]])

generated_words = []

for i in range(57):
    output = model(input_tensor)
    predicted_index = output.argmax(dim =-1)[0,1]
    predicted_word = id2word[predicted_index.item()]
    print(predicted_word, end = " ")
    generated_words.append(predicted_word)
    input_tensor = torch.cat([input_tensor, predicted_index.unsqueeze(0).
 ↪unsqueeze(0)],dim = -1)
    time.sleep(.5)
```

```
that shame shame that that that that fool that shame that that that shame that
shame fool that shame that that that that fool that that that shame that that
that that that that that fool that that that that fool fool shame fool that fool
tennessee that shame fool shame shame that fool that tennessee shame
```

[ ]:

7 6: PreTrained Decoder

[13]:
```
%pip install transformers sentencepiece accelerate

cache_dir = "./llm_architecture/deployment"
```

```
Requirement already satisfied: transformers in /opt/conda/lib/python3.11/site-
packages (5.2.0)
Requirement already satisfied: sentencepiece in /opt/conda/lib/python3.11/site-
packages (0.2.1)
Requirement already satisfied: accelerate in /opt/conda/lib/python3.11/site-
packages (1.12.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
/opt/conda/lib/python3.11/site-packages (from transformers) (1.4.1)
```

```
Requirement already satisfied: numpy>=1.17 in /opt/conda/lib/python3.11/site-
packages (from transformers) (2.0.1)
Requirement already satisfied: packaging>=20.0 in
/opt/conda/lib/python3.11/site-packages (from transformers) (24.1)
Requirement already satisfied: pyyaml>=5.1 in /opt/conda/lib/python3.11/site-
packages (from transformers) (6.0.2)
Requirement already satisfied: regex!=2019.12.17 in
/opt/conda/lib/python3.11/site-packages (from transformers) (2026.2.19)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
/opt/conda/lib/python3.11/site-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in /opt/conda/lib/python3.11/site-
packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in
/opt/conda/lib/python3.11/site-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /opt/conda/lib/python3.11/site-
packages (from transformers) (4.66.5)
Requirement already satisfied: psutil in /opt/conda/lib/python3.11/site-packages
(from accelerate) (6.0.0)
Requirement already satisfied: torch>=2.0.0 in /opt/conda/lib/python3.11/site-
packages (from accelerate) (2.10.0+cpu)
Requirement already satisfied: filelock in /opt/conda/lib/python3.11/site-
packages (from huggingface-hub<2.0,>=1.3.0->transformers) (3.20.0)
Requirement already satisfied: fsspec>=2023.5.0 in
/opt/conda/lib/python3.11/site-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (2023.6.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in
/opt/conda/lib/python3.11/site-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (1.2.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/opt/conda/lib/python3.11/site-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.27.0)
Requirement already satisfied: shellingham in /opt/conda/lib/python3.11/site-
packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.5.4)
Requirement already satisfied: typing-extensions>=4.1.0 in
/opt/conda/lib/python3.11/site-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (4.12.2)
Requirement already satisfied: sympy>=1.13.3 in /opt/conda/lib/python3.11/site-
packages (from torch>=2.0.0->accelerate) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in
/opt/conda/lib/python3.11/site-packages (from torch>=2.0.0->accelerate) (3.3)
Requirement already satisfied: jinja2 in /opt/conda/lib/python3.11/site-packages
(from torch>=2.0.0->accelerate) (3.1.4)
Requirement already satisfied: typer>=0.24.0 in /opt/conda/lib/python3.11/site-
packages (from typer-slim->transformers) (0.24.0)
Requirement already satisfied: anyio in /opt/conda/lib/python3.11/site-packages
(from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.4.0)
Requirement already satisfied: certifi in /opt/conda/lib/python3.11/site-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers)
```

```
(2024.12.14)
Requirement already satisfied: httpcore==1.* in /opt/conda/lib/python3.11/site-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers)
(1.0.5)
Requirement already satisfied: idna in /opt/conda/lib/python3.11/site-packages
(from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (3.8)
Requirement already satisfied: sniffio in /opt/conda/lib/python3.11/site-
packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers)
(1.3.1)
Requirement already satisfied: h11<0.15,>=0.13 in
/opt/conda/lib/python3.11/site-packages (from
httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers)
(0.14.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in
/opt/conda/lib/python3.11/site-packages (from
sympy>=1.13.3->torch>=2.0.0->accelerate) (1.3.0)
Requirement already satisfied: click>=8.2.1 in /opt/conda/lib/python3.11/site-
packages (from typer>=0.24.0->typer-slim->transformers) (8.3.1)
Requirement already satisfied: rich>=12.3.0 in /opt/conda/lib/python3.11/site-
packages (from typer>=0.24.0->typer-slim->transformers) (14.3.3)
Requirement already satisfied: annotated-doc>=0.0.2 in
/opt/conda/lib/python3.11/site-packages (from typer>=0.24.0->typer-
slim->transformers) (0.0.4)
Requirement already satisfied: MarkupSafe>=2.0 in
/opt/conda/lib/python3.11/site-packages (from jinja2->torch>=2.0.0->accelerate)
(2.1.5)
Requirement already satisfied: markdown-it-py>=2.2.0 in
/opt/conda/lib/python3.11/site-packages (from
rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/opt/conda/lib/python3.11/site-packages (from
rich>=12.3.0->typer>=0.24.0->typer-slim->transformers) (2.18.0)
Requirement already satisfied: mdurl~=0.1 in /opt/conda/lib/python3.11/site-
packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer>=0.24.0->typer-
slim->transformers) (0.1.2)
Note: you may need to restart the kernel to use updated packages.
```

[14]:
```python
from transformers import GPT2LMHeadModel, GPT2Tokenizer

tokenizer = GPT2Tokenizer.from_pretrained("gpt2")
model = GPT2LMHeadModel.from_pretrained("gpt2")
inputs = tokenizer("Hello world", return_tensors = "pt")
with torch.no_grad():
    out = model(**inputs)
```

```
Loading weights:   0%|          | 0/148 [00:00<?, ?it/s]
```

[15]:
```
# tokenizer_large = GPT2Tokenizer.from_pretrained("gpt2-XL", cache_dir =␣
 ↪cache_dir+"/models")
# model_large = GPT2LMHeadModel.from_pretrained("gpt2-XL", cache_dir =␣
 ↪cache_dir+"/models")
```

[28]:
```
def generate_text(prompt, cache_dir = cache_dir, model_size = "small",␣
 ↪max_tokens = 250, delay = .1):
    model_name = "gpt2"
    tokenizer = GPT2Tokenizer.from_pretrained(model_name, cache_dir␣
 ↪=f"{cache_dir}/models")
    model  = GPT2LMHeadModel.from_pretrained(model_name, cache_dir =␣
 ↪f"{cache_dir}/models")

    inputs = tokenizer.encode(prompt,return_tensors = 'pt')
    attention_mask = torch.ones(inputs.shape, dtype = torch.long)
    pad_token_id = tokenizer.eos_token_id

    print(prompt + "\n", end = " " , flush = True)
    for _ in range(max_tokens):
        outputs = model.generate(
            inputs,
            max_length = inputs.shape[-1] + 1,
            do_sample = True,
            pad_token_id = pad_token_id,
            attention_mask=attention_mask
        )
        generated_token = outputs[0][-1].unsqueeze(0).unsqueeze(0)
        generated_word = tokenizer.decode(generated_token[0])
        print(generated_word, end = " " , flush = True)
        inputs = torch.cat([inputs, generated_token],dim = -1)
        attention_mask = torch.cat([attention_mask, torch.ones((1,1), dtype =␣
 ↪torch.long)], dim = -1)
        time.sleep(delay)
```

[29]:
```
prompt = "it is always sunny in Philidelphia"

generate_text(prompt, max_tokens = 50)
```

```
Loading weights:   0%|          | 0/148 [00:00<?, ?it/s]
```

```
it is always sunny in Philidelphia
 ." For some a time the city sought a way to replace one of its
former buildings . They needed a place to park and use the beach
instead of a park , so they built a concrete deck in the old
parking lot around 10 p . m .
```